

Fig 3k. MLM (the only multivariate estimator) reshuffles BOTH axes (WT_heat)

Rank of each TF under the 6 decoupleR statistics; boxed MLM column is the outlier. Spearman vs ULM (full 658-TF ranking) is 0.62 for MLM but 0.97-1.00 for every univariate statistic. Under MLM, the univariate top set is demoted across BOTH axes: Rela 3->350, Nfkb1 6->86, Hif1a's collapse is ONE instance of this general MLM de-confounding reshuffle, not a HIF-specific quirk.



decoupleR statistic (rho = Spearman rank-corr vs ULM, full WT_heat ranking)